

Quick Activity 2: Canteen Group Order

A canteen sells rice meals at ₱85 each and drinks at ₱25 each. Every order receives a 10% discount on the rice-meal subtotal only. The canteen also charges a ₱5 packaging fee per rice meal.

Develop an algorithm that calculates the final bill and the customer's change.

Rules

- Drinks and packaging fees receive no discount.
- Add the packaging fee after calculating the discount.
- Assume quantities are positive whole numbers and payment is enough to cover the final bill.

Inputs:

- Number of rice meals
- Number of drinks
- Payment

Processes:

- Calculate rice-meal subtotal (rice meals $\times$ ₱85)
- Calculate drink subtotal (drinks $\times$ ₱25)
- Calculate 10% discount on rice-meal subtotal only
- Subtract discount from rice-meal subtotal
- Calculate packaging fee (rice meals $\times$ ₱5)
- Add packaging fee to discounted rice subtotal
- Calculate final bill (discounted rice + packaging fee + drink subtotal)
- Calculate change (payment – final bill)

Outputs:

- Final bill
- Change

Tasks:

- 1.) Input number of rice meals, drinks, and payment.
- 2.) Compute the rice-meal subtotal.
- 3.) Compute the drink subtotal.
- 4.) Compute the 10% discount.

- 5.) Subtract the discount from the rice-meal subtotal.
- 6.) Compute the packaging fee.
- 7.) Add the discounted meal subtotal, drink subtotal, and packaging fee.
- 8.) Subtract the final bill from the payment.
- 9.) Display the final bill and change.

PseudoCode

START

```

READ riceMeals, drinks, payment
mealSubtotal ← riceMeals × 85
drinkSubtotal ← drinks × 25
discount ← mealSubtotal × 0.10
discountedMealSubtotal ← mealSubtotal – discount
packagingFee ← riceMeals × 5
finalBill ← discountedMealSubtotal + drinkSubtotal + packagingFee
change ← payment – finalBill
DISPLAY finalBill
DISPLAY change

```

END

DRY RUN CALCULATIONS:

Test Case 1:

Rice Meals = 3
 Drinks = 4
 Payment = ₱500

Step	Variable	Calculation	C
1	mealSubtotal	3x ₱85	₱255.00
2	drinkSubtotal	4x ₱25	₱100.00
3	discount	₱255 x 0.10	₱25.50
4	discountedMealSubtotal	₱255- ₱25.50	₱229.50
5	packagingFee	3x ₱5	₱15.00
6	finalBill	₱229.50 + ₱100 + ₱15	₱344.50
7	change	₱500 – ₱344.50	₱155.50

Final bill: ₱344.50

Change: ₱155.50

Test Case 2:

Rice Meals = 5

Drinks = 3

Payment = ₱600

Step	Variable	Calculation	Result
1	mealSubtotal	5x ₱85	₱425.00
2	drinkSubtotal	3x ₱25	₱75.00
3	discount	₱425 x 0.10	₱42.50
4	discountedMealSubtotal	₱425 - ₱42.50	₱382.50
5	packagingFee	5 × ₱5	₱25.00
6	finalBill	₱382.50 + ₱75 + ₱25	₱482.50
7	change	₱600 - ₱482.50	₱117.50

Final bill: **₱482.50**

Change: **₱117.50**

Explain: Why would applying the 10% discount to the entire bill produce an incorrect result?

Because the 10% discount applies on the rice-meal subtotal only and disregards the drinks and packaging fee.